

EVNet Sentinel — Runbook

Starting and stopping every server

Run all commands from the repository root unless a step says otherwise.

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1 What there is to run

Server	Start	Address	Needs
Website (demo)	<code>make web</code>	<code>localhost:3100</code>	Node.js only
Website (live reload)	<code>make web-dev</code>	<code>localhost:3000</code>	Node.js only
Prediction API	<code>make api</code>	<code>localhost:8000</code>	Python and the trained models

The website and the API are independent. **The website does not call the API:** it reads results already exported to `frontend/data/*.json`, so it runs on a fresh clone with no dataset and no Python. The API classifies one network flow per request. No database or other service needs to be running.

2 One-time setup

Tool	Version
Node.js	20.9 or newer (developed on 26.4)
Python	3.10 or newer (developed on 3.13)
GNU Make	any

```
make setup          # Python packages (requirements.txt)
cd frontend && npm install  # website packages
cd ..
```

3 Website

3.1 Start

For a demo, use the production build. It is faster, and it always shows the latest code.

```
make web
```

This runs `npm run build`, then `next start -p 3100`. The build takes about 30 seconds; wait for Ready, then open <http://localhost:3100>.

While editing the frontend, use live reload instead:

```
make web-dev
```

3.2 Demo route

Page	What to show
http://localhost:3100	<i>Beat the Sentinel</i> : drag an attack onto a station. <i>Tune the detector</i> : drag the confidence bar. Then scroll the Mumbai story.
http://localhost:3100/simulation	Pick an attack, choose <i>Sent from</i> , launch; or run a campaign and read the incident log.
http://localhost:3100/dashboard	<i>Attack insights</i> tab; the confusion matrix and drift timeline on <i>Findings</i> .
http://localhost:3100/docs	The “Method at a glance” pipeline and the architecture.

3.3 Stop

Press **Ctrl+C** in the terminal running it. If that terminal is gone, stop whatever holds the port:

```
lsof -ti:3100 | xargs kill  # production website
lsof -ti:3000 | xargs kill  # live-reload website
```

4 Prediction API

4.1 Start

```
make api
```

This serves the **corrected (leak-free) models** produced by `src/evaluation/run_experiments.py` — Random Forest, Decision Tree, Logistic Regression and SVM from `saved_models/multiseed/seed-42/` — with that run's scaler. It is ready in about 10 seconds. For another training seed: `make api API_SEED=1337` (or 2024).

4.2 Check that it works

```
curl http://localhost:8000/
```

The reply lists `models_loaded` and should contain `"scaler_loaded": true`. Classify a real held-out flow:

```
curl -X POST http://localhost:8000/predict \  
-H "Content-Type: application/json" \  
-d @docs/examples/predict_syn_flood.json
```

Expected: `{"prediction_class": "SYN_Flood", "status": "success"}`. The second sample, `predict_port_scan.json`, returns `Slowloris_Scan`. That is the scan confusion described in the findings, not a fault. Interactive API documentation is at <http://localhost:8000/docs>.

A request names its model and sends the 60 flow features. Always pass `model_name` (one of those listed by `GET /`): the default, `arfadwin_model`, does not exist. Missing features are filled with 0, so send all 60 for a meaningful answer.

4.3 Stop

Press `Ctrl+C`, or:

```
lsof -ti:8000 | xargs kill
```

4.4 First run on a fresh clone

The model files are not in git. Generate them once. This takes about 9 minutes and needs the raw CICEVSE2024 captures:

```
python3 src/evaluation/run_experiments.py --raw-dir <raw-captures> \  
--feature-set extended --seeds 42 1337 2024
```

This writes `saved_models/multiseed/` and `data/multiseed/`, which `make api` reads.

Do not use `make run` or the Docker image for a demo

Both load the older artifacts in the root of `saved_models/`, including a scaler fitted *with* the six leaked timestamp columns. Requests carrying the corrected 60 features then fail with “Feature names seen at fit time, yet now missing: `bidirectional_first_seen_ms...`”. Use `make api`.

5 Everything at once

5.1 Start

Two terminals:

```
make web      # terminal 1: website on :3100
make api      # terminal 2: API on :8000
```

5.2 Stop

Ctrl+C in each terminal, or all servers at once:

```
for p in 3000 3100 8000; do lsof -ti:$p | xargs kill 2>/dev/null; done
```

5.3 Confirm nothing is left running

```
lsof -iTCP:3000 -iTCP:3100 -iTCP:8000 -sTCP:LISTEN
```

No output means every server is stopped.

6 Refreshing what the website shows

The website’s numbers come from committed JSON in `frontend/data/`. These commands are only needed after retraining or changing the evaluation; each needs the local models and data above. Run `make web` again afterwards.

Command	Regenerates
<code>make insights</code>	Per-attack tests and the <i>Attack insights</i> tab (<code>insights.json</code>)
<code>make figures</code>	Confusion, separability, drift (<code>paper.json</code>); the hero's tuning data (<code>hero.json</code>); paper figures in <code>docs/figures/</code>
<code>src/evaluation/export_dashboard_data.py</code>	Findings and replay (<code>findings.json</code> , <code>simulation.json</code>)
<code>src/evaluation/export_simulation_data.py</code>	The simulation console's flow pool (<code>network-sim.json</code>)
<code>src/reproduction/attack_routes.py</code>	Attacker and target of every capture (<code>attack-routes.json</code>); needs the raw captures

7 Troubleshooting

Symptom	Fix
"Port 3100 is in use" (or 3000, 8000)	A previous server is still running: <code>lsof -ti:3100 xargs kill</code> .
Website shows old content after an edit	Dev and production share one build folder. Stop the server and run <code>make web</code> .
API: "No models are loaded"	Models missing: run <code>run_experiments.py</code> (Section 4.4).
API: "Model ... not found"	Pass a <code>model_name</code> listed by <code>GET /</code> .
API: "..._seen_ms" missing	Started with <code>make run</code> ; use <code>make api</code> .
npm complains about Node	Install Node.js 20.9 or newer.
Landing story does not animate	The viewer's system has <i>Reduce motion</i> on; the page shows the story as text by design.